

Joshua Aney

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EDUCATION

Montana State University

Bachelor of Science, Computer Science, GPA: 3.96

Bozeman, MT

May 2026

HONORS & AWARDS

President's List: Spring 2022; Spring 2023, Fall 2023; Spring 2024; Spring 2025, Fall 2025; Spring 2026, Fall 2026

Dean's List: Fall 2022; Fall 2024

RESEARCH EXPERIENCE

Research Technologist

CaMP Lab (Cybernetics and Motor Physiology), Mayo Clinic

June 2026 – Present

Rochester, MN

- **Tools:** MATLAB, Signal Processing, Adobe Illustrator, Linear Algebra, Technical Writing
- Previously Summer Research Intern in the same lab, 2024 and 2025.
- Conducted algorithmic analysis on human sEEG data to study how the brain responds to auditory stimuli.
- Implemented signal processing pipelines including noise reduction, baseline correction, and artifact rejection.
- Applied and modified Canonical Response Parameterization to allow for better temporal feature extraction when applied to auditory evoked responses.
- Created a new method for dynamic brain network identification using evoked responses.
- Leading development of a reference and preprocessing guide for general sEEG data analysis.

Undergraduate Research Assistant

HACR Lab (Harnessing Automation in Cybersecurity Reasoning), Montana State University

January 2026 – June 2026

Bozeman, MT

- **Tools:** PyTorch, Ghidra, High Performance Computing, SLURM
- Built pipelines to produce control flow graphs from compiled binary packages for code embedding generation.
- Designed Graph Neural Network architectures using PyTorch based on recent research publications.
- Ran machine learning workloads on Tempest, Montana State University's high performance computing system.

TEACHING & CURRICULUM DEVELOPMENT

Independent Study – CSCI 491-7 Hypermedia Systems

Montana State University

Fall 2025

Bozeman, MT

- **Tools:** Java, SQLite, HTMX, HTML/CSS, MVC Architecture, Database Design
- Collaborated with a professor and a team of three students to design and develop a semester-long project template for an advanced web development course teaching a hypermedia-first approach using server-side rendering with HTMX.
- Implemented core website functionality, including the MVC controller layer, authentication and authorization systems, and multi-step business workflows, for a veterinary clinic management system.
- Designed a database schema with 11 interconnected tables, including polymorphic relationships and conversation-based messaging, to teach role-based access control, pagination, and transactional integrity.
- Structured a progressive assignment sequence so students could implement new controllers by following established patterns in the codebase.

PUBLICATIONS & PRESENTATIONS

Papers in Preparation

- Aney, J., et al. *Response Lag Correlation: Identifying response networks through evoked response lags.*
- Aney, J., et al. *A Reference and Preprocessing Guide for sEEG Data.*

Conference Abstracts

- Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects*. BRAIN Initiative Conference, 2026.
- Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects*. GATTLE, Mayo Clinic, 2026.

Posters

- Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects*. Poster presented at BRAIN Initiative Conference, 2026.

SKILLS

Languages: Python, MATLAB, Java, C, SQL, C#

Frameworks: PyTorch, TensorFlow, WebDataset, .NET Core, SLURM, NumPy, Matplotlib

Tools: Linux, JetBrains Products, GitHub, VSCode, Adobe Illustrator, Overleaf, Claude Code

TECHNICAL PROJECTS

- Machine Learning and AI Models from Scratch** | *Python, NumPy, Linear Algebra, Latex* 2024-2026
- Implemented the following machine learning algorithms from scratch: Naïve Bayes, K-Nearest Neighbors, K-Means, and DBSCAN, Multi-Layer-Perceptron, Q-Learning, SARSA, and Value Iteration
 - Implemented a First-Order-Logic engine to perform inference on knowledge bases using resolution.
 - Implemented the following optimization algorithms from scratch: Backpropagation, Genetic Algorithm, Differential Evolution, Particle Swarm Optimization, and Simulated Annealing
- Prize-Collecting TSP Solver** | *Python, CPLEX, Linear Programming, Optimization Algorithms* 2026
- Formulated the Prize-Collecting Traveling Salesman Problem as a mixed-integer linear program and solved it exactly using IBM CPLEX.
 - Implemented a genetic algorithm as an alternative heuristic solver and compared solution quality and runtime against the exact CPLEX formulation.
- Bozeman Bike Trails Application** | *C#, Razor Pages, SQL Server, Mapbox* 2024
- Developed a web application to track user-ridden bike trail GPS data
- Hand Tracking / ASL Alphabet Detection** | *Python, TensorFlow, MediaPipe* 2024
- Created a pipeline to collect data, train a dense neural network, and provide real time predictions for video inputs.

OTHER ACTIVITIES

- President** August 2023 – May 2026
Montana State University Tennis Club Bozeman, MT
- Section Leader** April 2025 – January 2026
Montana State University Drumline Bozeman, MT
- Tennis Coach** Summers 2023, 2025
Rochester Tennis Connection Rochester, MN
- Substitute Tennis Coach** 2022 – 2026
Bobcat Anderson Tennis Center, Montana State University Bozeman, MT